

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION | COURSE CODE: ITA05
COURSE OUTCOME COVERED: CO4 (BTL 5) | WEIGHTAGE: 20%

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

9. Mobile Face Detection Trade-off

Constraints:

- Real-time operation required
- Accuracy $\geq 85\%$

Options:

- Viola-Jones: 80%, fast
- Deep Learning: 95%, slow

Compare and evaluate which approach should be chosen. Justify using constraint satisfaction logic.

ANSWER:

Requirements & Constraints: Real-time operation required, Accuracy $\geq 85\%$.

Option Evaluation:

- **Viola-Jones:** Fast (real-time speed), but achieves only 80% accuracy ($< 85\%$). Violates the mandatory accuracy constraint.
- **Deep Learning (Standard Heavy Model):** Achieves 95% accuracy ($\geq 85\%$), but processing speed is slow on standard mobile hardware. Violates the strict real-time execution constraint.

Decision: Neither off-the-shelf option satisfies both operational constraints simultaneously.

Justification: Viola-Jones violates accuracy thresholds, while standard Deep Learning violates real-time execution bounds. Selecting Viola-Jones risks false detections, whereas deploying standard Deep Learning causes lag and dropped frames.

Recommended Engineering Solution: Deploy a Lightweight / Quantized Deep Learning Architecture (such as MobileNet-SSD, BlazeFace, or ONNX-quantized models). This optimized hybrid approach achieves $\geq 85\%$ accuracy while leveraging hardware acceleration (Mobile NPU/GPU) to satisfy the real-time constraint.

Option / Model	Accuracy	Real-time Execution	Constraint Verification	Decision Status
Viola-Jones	80%	Pass (Fast)	Accuracy $< 85\%$	Rejected
Deep Learning (Heavy)	95%	Fail (Slow)	Latency Constraint Violated	Rejected
Lightweight DL (BlazeFace)	88% - 92%	Pass (NPU Accelerated)	All Constraints Satisfied	Selected / Recommended

16. Autonomous Navigation Constraint

System must:

- Accuracy $\geq 95\%$
- Latency < 70 ms

Options:

- Model A: 94%, 60 ms
- Model B: 96%, 80 ms
- Model C: 95%, 65 ms

Compare and evaluate. Justify your decision.

ANSWER:

Requirements & Constraints: Accuracy $\geq 95\%$, Latency < 70 ms (Hard real-time boundary for autonomous safety).

Option Evaluation:

- **Model A:** Latency = 60 ms (< 70 ms), but Accuracy = 94% ($< 95\%$). Fails safety accuracy floor.
- **Model B:** Accuracy = 96% ($\geq 95\%$), but Latency = 80 ms (> 70 ms). Fails latency requirement, introducing dangerous collision risks.
- **Model C:** Accuracy = 95% ($\geq 95\%$), Latency = 65 ms (< 70 ms). Meets both strict constraints simultaneously.

Decision: Select Model C.

Justification: Model C is the only candidate that satisfies all constraints without compromising safety. Model A violates the strict 95% accuracy requirement, making object detection unreliable. Model B violates the 70 ms response boundary, increasing braking distance and reaction delay. Model C hits the exact 95% accuracy baseline while operating safely within the 65 ms latency window.

Model	Accuracy ($\geq 95\%$)	Latency (< 70 ms)	Constraint Verification	Decision Status
Model A	94%	60 ms	Accuracy $< 95\%$ (Violates Safety Floor)	Rejected
Model B	96%	80 ms	Latency > 70 ms (Violates Time Bound)	Rejected
Model C	95%	65 ms	All Conditions Satisfied	Selected

20. Smart Parking System

Requirements:

- Detect vehicles accurately
- Low computational cost

Options:

- Viola-Jones: low cost, low accuracy
- YOLO: moderate cost, high accuracy

Compare and evaluate. Justify your decision.

ANSWER:

Requirements & Constraints: Accurate vehicle detection, Low computational cost for deployment on localized edge nodes.

Option Evaluation:

- **Viola-Jones:** Provides low computational overhead, but suffers from low detection accuracy, failing to reliably detect cars under severe angles, occlusions, or lighting changes.
- **YOLO (You Only Look Once):** Delivers high detection accuracy under diverse environmental variations (day/night, angles) with moderate computational requirements supported by modern edge processors.

Decision: Select YOLO (specifically Tiny-YOLO or YOLO Edge variants).

Justification: Smart parking systems require reliable vehicle identification to prevent incorrect slot allocation and false availability status. Viola-Jones relies on rigid Haar features that frequently fail on varied car shapes, shadows, and partial occlusions. YOLO provides robust deep feature extraction. Furthermore, lightweight YOLO variants (e.g., YOLOv8n / Tiny-YOLO) run efficiently on low-cost edge hardware (such as Jetson Nano or Raspberry Pi with AI accelerators), fully satisfying computational constraints while ensuring high accuracy.

Method / Model	Detection Accuracy	Computational Cost	Constraint Satisfaction	Decision Status
Viola-Jones	Low (Unreliable)	Very Low	Fails Accuracy Requirement	Rejected
YOLO (Edge Variant)	High (Robust)	Moderate / Low (Optimized)	Satisfies Accuracy & Hardware Bounds	Selected

8. AR Motion Tracking Constraint Problem

An AR system must:

- Maintain stable tracking (<5% drift)
- Run on mobile hardware

Options:

- Optical Flow: detailed but computationally intensive
- Motion Estimation: less detailed but efficient

Compare and evaluate which method satisfies both constraints. Justify your decision logically.

ANSWER:

Requirements & Constraints: Stable motion tracking with drift < 5%, Execution on mobile hardware resource budgets.

Option Evaluation:

- **Optical Flow (Dense):** Generates dense, highly detailed motion vectors but requires intensive per-pixel computation. This causes thermal throttling, rapid battery drain, and frame rate drops on mobile hardware, ultimately inducing tracking lag.
- **Motion Estimation / Sparse Tracking:** Provides block-based or sparse feature tracking. Highly efficient computationally, enabling real-time operation on mobile devices while maintaining low tracking drift when combined with inertial sensors (VIO).

Decision: Select Motion Estimation (or Sparse Optical Flow / Feature-based Tracking).

Justification: Dense Optical Flow overloads mobile processors, causing system jitter and frame drops that exceed the <5% drift limit. Motion Estimation / Sparse Tracking reduces computational load while keeping spatial drift within the 5% threshold when paired with mobile IMUs (Visual-Inertial Odometry), maintaining a fluid augmented reality overlay.

Tracking Method	Drift Stability (<5%)	Mobile Compute Load	Constraint Verification	Decision Status
Dense Optical Flow	High (prone to lag)	Intensive (Thermal Overload)	Violates Mobile Hardware Limits	Rejected
Motion Estimation (Sparse)	Satisfies (< 5% Drift)	Low / Efficient	Satisfies Both Constraints	Selected

22. Edge Device Vision System

Constraints:

- Low power
- Real-time performance

Options:

- Deep Learning: accurate, heavy
- Lightweight Model: moderate accuracy, efficient

Compare and evaluate. Justify logically.

ANSWER:

Requirements & Constraints: Low power consumption, Real-time execution on edge hardware.

Option Evaluation:

- **Deep Learning (Standard Heavy):** High accuracy, but heavy compute requirements (> 10 W power consumption), high thermal output, and non-real-time execution on low-power edge platforms. Fails both power and latency constraints.
- **Lightweight Model:** Moderate accuracy, high computational efficiency, low power consumption (< 2-5 W), real-time response capability. Meets all operational edge bounds.

Decision: Select Lightweight Model.

Justification: Edge deployments are strictly bound by thermal dissipation limits and battery/power budgets. Standard deep learning architectures draw excessive power and trigger thermal throttling on edge devices, leading to frame dropping. Lightweight models

(e.g., MobileNet, SqueezeNet, Pruned CNNs) fit within power/thermal budgets while delivering real-time inferences, making them the only feasible deployment choice.

Model Type	Power Consumption	Real-Time Speed	Compute Footprint	Decision Status
Standard Heavy DL	High (> 10 W)	Non-Real-Time on Edge	Heavy Compute	Rejected
Lightweight Model	Low (< 3 W)	Real-Time (Pass)	Efficient Footprint	Selected

18. Crowd Monitoring System

Requirements:

- Stable detection in dense crowds
- Moderate accuracy acceptable

Options:

- Optical Flow: unstable in dense scenes
- Motion Models: stable

Compare and evaluate. Justify your decision.

ANSWER:

Requirements & Constraints: Stable tracking/detection in dense crowd conditions, Moderate accuracy baseline acceptable.

Option Evaluation:

- **Optical Flow:** Calculates fine pixel-level vectors, but suffers severe tracking failure, vector noise, and trajectory corruption due to vehicle/person occlusion in dense crowds.
- **Motion Models:** Uses macro-level visual features, motion density distributions, or block motion vectors. Highly stable in dense crowd conditions with acceptable moderate accuracy.

Decision: Select Motion Models.

Justification: In high-density crowds, individual pixel trajectories overlap significantly, creating optical flow aperture problems and noise. Motion Models treat crowd dynamics as macro flows or region-based movements, providing consistent stability. Because the system explicitly accepts moderate accuracy, Motion Models meet the exact functional requirements while avoiding tracking collapse.

Approach	Dense Crowd Stability	Accuracy Level	Constraint Verification	Decision Status
Optical Flow	Unstable (Severe Noise)	High Fine Detail / Low Stability	Fails Crowd Stability Constraint	Rejected
Motion Models	High (Stable Macro Tracking)	Moderate (Acceptable)	All Requirements Satisfied	Selected

30. Motion Animation System

Requirements:

- Smooth motion
- Real-time rendering

Options:

- Motion Models: realistic, slow
- Motion Estimation: fast, less realistic

Compare and evaluate design approach. Justify your decision.

ANSWER:

Requirements & Constraints: Smooth visual motion, Real-time rendering speed (e.g., ≥ 30 -60 FPS).

Option Evaluation:

- **Motion Models (Physics-Based):** Generates physically accurate and highly realistic animation, but incurs high computational costs and slow rendering times, failing real-time limits.
- **Motion Estimation / Kinematic Models:** Provides fast kinematic approximations and motion interpolation. Ensures rapid frame generation and smooth visual transitions under tight frame time constraints.

Decision: Select Motion Estimation for interactive real-time gameplay/rendering, or adopt a Hybrid Architecture.

Justification: Real-time systems must render each frame within 16–33 ms. Full physics-based Motion Models exceed this budget, causing rendering lag. Motion Estimation meets real-time frame rates while ensuring smooth visual motion. For hybrid designs, Motion Estimation handles interactive gameplay, whereas full Motion Models are reserved for pre-rendered cutscenes.

Design Approach	Motion Smoothness	Rendering Speed	Real-Time Suitability	Decision Status
Physics-Based Motion Models	High (Realistic)	Slow (> 33 ms/frame)	Unsuitable for Interactive Real-Time	Rejected (Gameplay)
Motion Estimation / Kinematics	Good (Smooth)	Fast (< 16 ms/frame)	Fully Satisfies Real-Time Bounds	Selected

5. Depth Estimation Strategy

Constraints:

- Only one camera available
- Moderate depth accuracy acceptable

Options:

- Stereo Vision: high accuracy, needs two cameras
- Structure from Motion: moderate accuracy, single camera

Compare both approaches and evaluate which is feasible. Justify using constraint-based reasoning.

ANSWER:

Requirements & Constraints: Single camera hardware constraint, Moderate depth accuracy acceptable.

Option Evaluation:

- **Stereo Vision:** Provides high depth accuracy, but strictly requires a dual-camera setup with calibrated baseline distance. Violates the single-camera hardware constraint.
- **Structure from Motion (SfM) / Monocular Depth:** Estimates 3D depth from sequential camera movement using a single lens. Provides moderate depth accuracy.

Decision: Select Structure from Motion (SfM).

Justification: Stereo Vision is physically impossible under the strict single-camera constraint. SfM reconstructs 3D spatial depth by tracking feature correspondences across moving single-camera frames. Since moderate depth accuracy is explicitly acceptable, SfM fulfills both the hardware limits and functional depth requirements.

Approach	Camera Requirement	Depth Accuracy	Constraint Verification	Decision Status
Stereo Vision	Dual Cameras Required	High	Violates Single Camera Constraint	Rejected
Structure from Motion (SfM)	Single Camera	Moderate	Satisfies All Hardware & Depth Bounds	Selected

Code of Conduct:

I DARANI E (192312120) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation